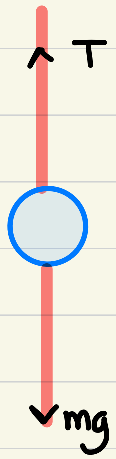
Newton's Laws of Motion

CONNECTED PARTICLES



Rest

$T = mg$

$a = 0$

$\uparrow$ $\boxed{\text{const vel}}$. $\downarrow$

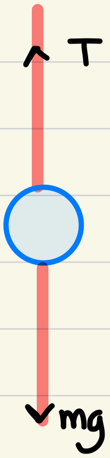
$T = mg$

UP $\quad T - mg = m(0)$

dN $\quad mg - T = m(0)$

Accelerate

✓ $F = ma$ ✓ Might is Right!

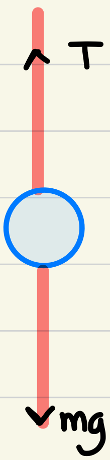


$\uparrow a$ UP

+ve −ve

$T > mg$ ignore

$\boxed{T - mg = ma}$ ✓



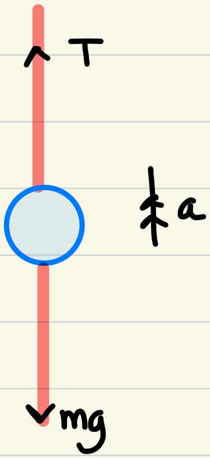
$\downarrow a$ $\boxed{\text{DOWN}}$

+ve −ve

$mg > T$

$\boxed{mg - T = ma}$

$F = ma$, UP



$m = 2\ kg \qquad g = 10\ m/s^2$

Case I : UP $\quad 1\ m/s^2$

$$T - mg = ma$$

$$T - 2(10) = 2(1)$$

$$T = 22N.$$

Case : UP $\quad -1\ m/s^2$

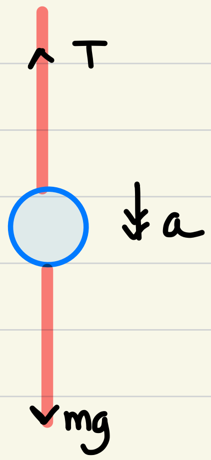
$$T - mg = ma$$

$$T - 2(10) = 2(-1)$$

$$T - 20 = -2$$

$$T = 18N$$

$F = ma$, DOWN



Case I : $a = 5\ m/s^2$

$$mg - T = ma$$

$$2(10) - T = 2(5)$$

$$20 - T = 10 \qquad \therefore\ T = 10\,N.$$

Case II : $a = -5\ m/s^2$

$$mg - T = ma$$

$$2(10) - T = 2(-5)$$

$$20 - T = -10$$

$$T = 30\,N.$$

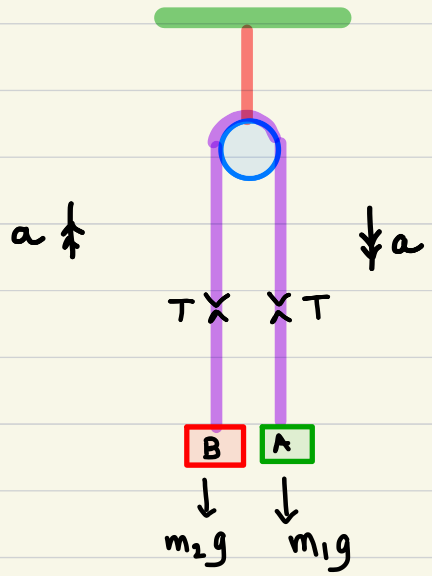
CONNECTED PARTICLES

- ✓ particles
- ✓ system
- ✓ common acceleration ⇒ accn. of the system.
  As long as string is taut
- ✓ one string ⟶ one tension

  Two strings ⟶ Two Tension
- ✓ Pulleys → smooth, → fixed
- ✓ string → inextensible, → Light

Pulley Systems

Case I : Simple Pulley



$m_1 = m_2 \rightarrow$ system remains at rest.

$m_1 > m_2$

particle A : $m_1g - \cancel{T} = m_1a$

particle B : $\cancel{T} - m_2g = m_2a$

system : $m_1g \boxed{-} m_2g = m_1a + m_2a$

$= \underline{(m_1+m_2)}a$

$m_1 = 0.6\,kg \quad m_2 = 0.4\,kg \quad g = 10\,m/s^2$

$0.6g - T = 0.6a$

$T - 0.4g = 0.4a$

$0.6g - 0.4g = (0.6+0.4)a$

$0.2g = 1.0a$

$0.2(10) = 1.0a$

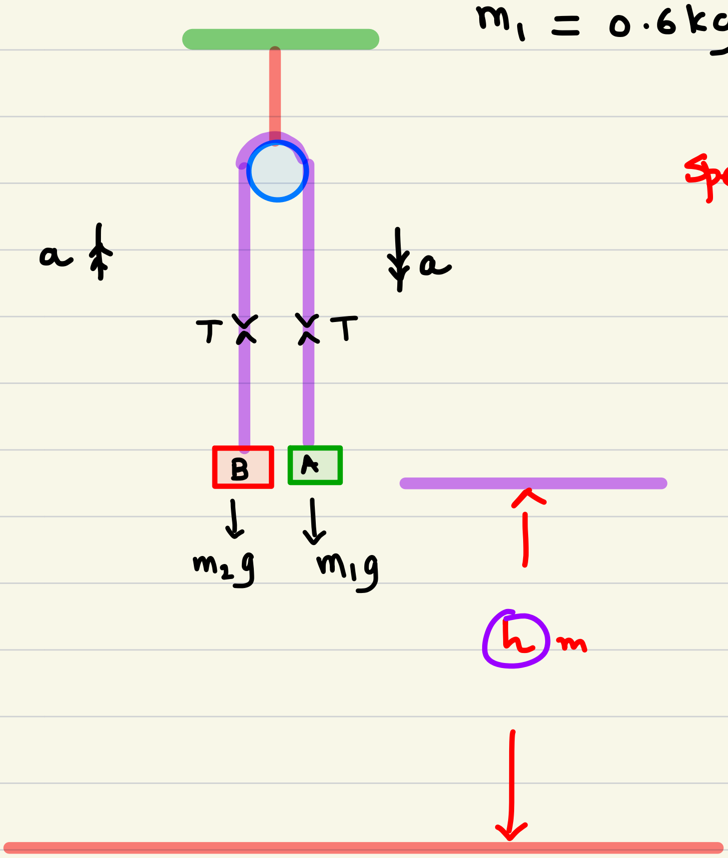
$\underline{a = 2\,m/s^2}$

$T - 0.4(10) = 0.4(2)$

$T - 4 = 0.8$

$\underline{T = 4.8\,N}$

$m_1 = 0.6\,kg \qquad m_2 = 0.4\,kg \qquad g = 10\,m/s^2$



Speed with which the heavier particle hits the floor.

$u = 0$ (Rest)

$a = 2\ m/s^2$

$s = 1\ m$

$v^2 = u^2 + 2as$

$= 0^2 + 2(2)(1)$

$v = 2\ m/s$ ✓

$v = u + at$

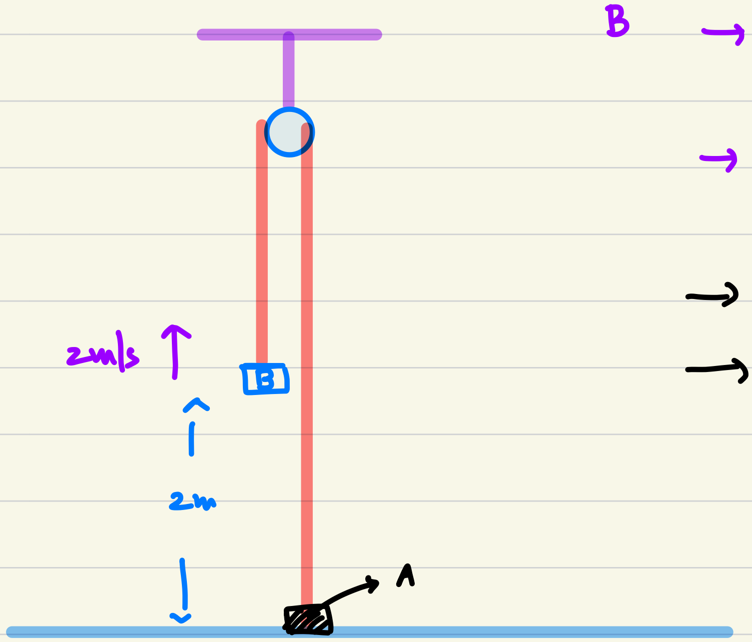
$2 = 0 + 2(t) \quad \therefore t = 1\ sec.$

Max height above ground reached by the lighter particle (B).

particle A $\rightarrow$ hits the floor $\rightarrow$ 2 m/s

B $\rightarrow$ is Now 2m above the ground.

2m/s.



B → free particle

→ freely moving under gravity

→ string slack

→ No Tension in the string.

$u = 2\,m/s$

$v = 0$

$a = -10\,m/s^2$.

$v = u + at$

$0 = 2 + (-10)t$

$t = 0.2\,sec$

$v^2 = u^2 + 2as$

$0^2 = 2^2 + 2(-10)s$

$\frac{4}{20} = s$ OR $s = 0.2\,m$

Max ht $\underbrace{1m + 1m}_{sys} + \underbrace{0.2m}_{free}$

$= 2.2\,m$.

Time to reach max ht from the moment it was released

$\underbrace{1.0\,sec}_{system} + \underbrace{0.2\,sec}_{free}$

$= 1.2\,sec$

Taut

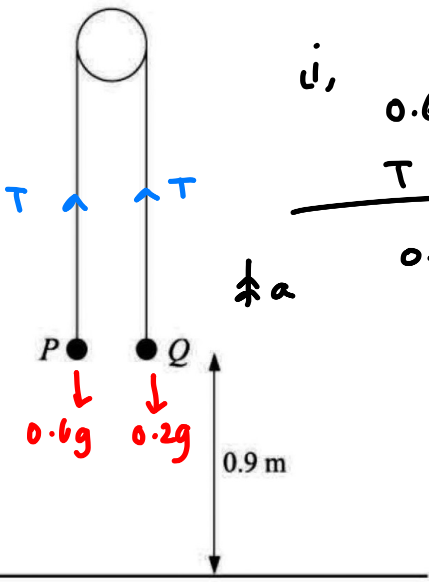
Lec #01

N. Laws ....

"Connected Particles"

case I $\longrightarrow$ simple Pulley

Question 6:



(ii) $u = 0$

$a = 5$

$s = 0.9$

$v^2 = u^2 + 2as$

$= 0^2 + 2(5)(0.9)$

$v = 3 \text{ m/s}$

$v = u + at$

$3 = 0 + (5)t \quad \therefore t = 0.6 \text{ sec}$

(i)

$0.6g - T = 0.6a$

$T - 0.2g = 0.2a$

$0.6g - 0.2g = 0.8a$

$0.4(10) = 0.8a$

$a = 5 \text{ m/s}^2$

$T = 0.2g + 0.2a$

$= 0.2[10 + 5]$

$T = 3 \text{ N}$

Particles P and Q, of masses 0.6 kg and 0.2 kg respectively, are attached to the ends of a light inextensible string which passes over a smooth fixed peg. The particles are held at rest with the string taut. Both particles are at a height of 0.9 m above the ground (see diagram). The system is released and each of the particles moves vertically. Find

(i) the acceleration of P and the tension in the string before P reaches the ground, [5]

(ii) the time taken for P to reach the ground. [2]

June 2007 Paper 4 Q4

Question 9:

i, $0.5g - T = 0.5a \rightarrow$ ①

$T - mg = ma \rightarrow$ ②

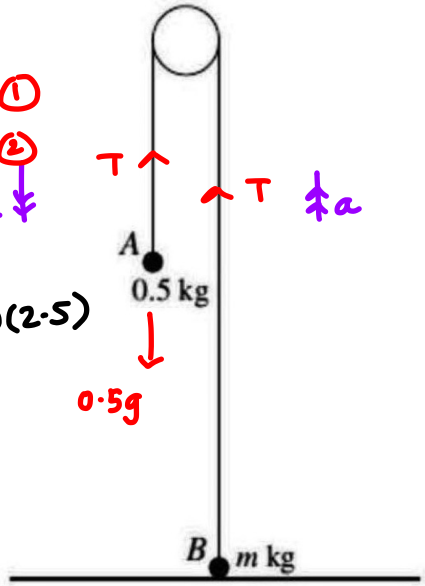
$0.5g - mg = (0.5+m)a$

$0.5(10) - m(10) = (0.5+m)(2.5)$

$5 - 10m = 1.25 + 2.5m$

$3.75 = 12.5m$

$m =$



$u = 0$

$a = ?$

$t = 2$ sec

$v = 5$ m/s

$v = u + at$

$5 = 0 + a(2)$

$a = 2.5$ m/s

$\boxed{1}$ $0.5(10) - T = 0.5(2.5)$

$5 - 1.25 = T$

$T = 3.75$N

Particles A and B, of masses 0.5 kg and m kg respectively, are attached to the ends of a light inextensible string which passes over a smooth fixed pulley. Particle B is held at rest on the horizontal floor and particle A hangs in equilibrium (see diagram). B is released and each particle starts to move vertically. A hits the floor 2 s after B is released. The speed of each particle when A hits the floor is $5\,\mathrm{m\,s^{-1}}$.

(i) For the motion while A is moving downwards, find

(a) the acceleration of A, ✓ [2]

(b) the tension in the string. ✓ [3]

(ii) Find the value of m. [3]

(iv) $T=0$, $-mg = ma$ $\therefore a = -g$

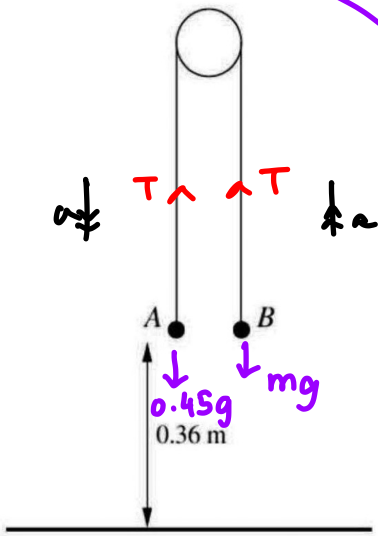
Question 10:

(i) $u=0$
$t = 0.6\,sec$
$s = 0.36\,m$

$s = ut + \frac{1}{2}at^2$

$0.36 = \frac{1}{2}(a)(0.6)^2$

$a = 2\,m/s^2$



$0.45g - T = 0.45a \rightarrow \boxed{A}$

$T - mg = ma \rightarrow \boxed{B}$

$0.45g - mg = (0.45+m)a$

(ii) $\boxed{A}$ $0.45g - T = 0.45a$

$0.45g - 0.45a = T$

$0.45[10-2] = T$

$T = 3.6\,N$

(iii) $\boxed{B}$ $T - mg = ma$

$T = mg + ma$

$T = m(g+a)$

$3.6 = m(10+2)$

$m = 0.3\,kg.$

Particles A and B are attached to the ends of a light inextensible string which passes over a smooth pulley. The system is held at rest with the string taut and its straight parts vertical. Both particles are at a height of 0.36 m above the floor (see diagram). The system is released and A begins to fall, reaching the floor after 0.6 s.

(i) Find the acceleration of A as it falls. [2]

The mass of A is 0.45 kg. Find

(ii) the tension in the string while A is falling, [2]

(iii) the mass of B, [3]

(iv) the maximum height above the floor reached by B. [3]

June 2009 Paper 4 Q6

(iv) free particle (A hits the floor)
does not rebound
String becomes slack
No more Tension
No more system

$\boxed{B}$ $u = 2\,m/s$

$v = 0$

$a = -10\,m/s^2$

$v^2 = u^2 + 2as$

$0^2 = 2^2 + 2(-10)s$

$s = 0.2\,m$

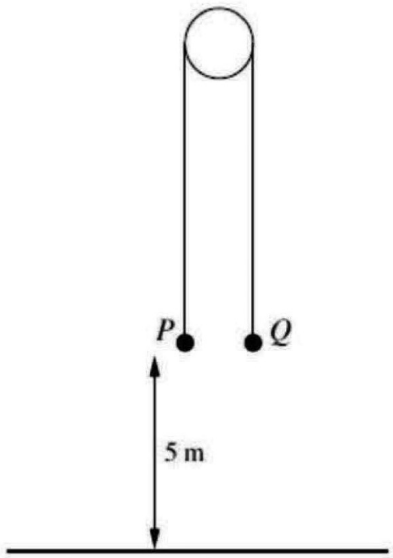
Max Height

$\underset{system}{0.36 + 0.36} + \underset{free}{0.2}$

$= 0.92\,m.$

Lec#02
LAWS OF MOTION

Question 11:



Particles P and Q, of masses 0.55 kg and 0.45 kg respectively, are attached to the ends of a light inextensible string which passes over a smooth fixed pulley. The particles are held at rest with the string taut and its straight parts vertical. Both particles are at a height of 5 m above the ground (see diagram). The system is released.

(i) Find the acceleration with which P starts to move. [3]

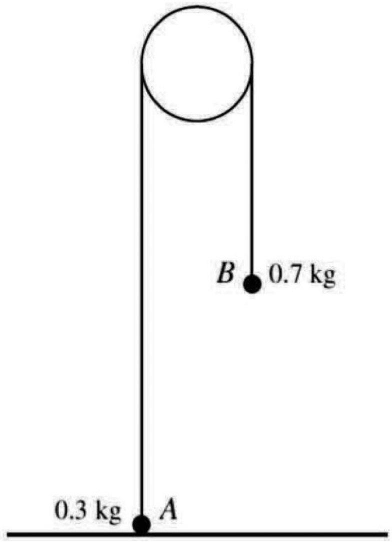
The string breaks after 2 s and in the subsequent motion P and Q move vertically under gravity.

(ii) At the instant that the string breaks, find

(a) the height above the ground of P and of Q, [2]

(b) the speed of the particles. [1]

Question 12:



Particles A and B, of masses 0.3 kg and 0.7 kg respectively, are attached to the ends of a light inextensible string which passes over a smooth fixed pulley. Particle A is held on the horizontal floor and particle B hangs in equilibrium. Particle A is released and both particles start to move vertically.

(i) Find the acceleration of the particles. [3]

The speed of the particles immediately before B hits the floor is $1.6\,\mathrm{m\,s^{-1}}$. Given that B does not rebound upwards, find

(ii) the maximum height above the floor reached by A, [3]

(iii) the time taken by A, from leaving the floor, to reach this maximum height. [3]

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✓ Pulley system → case I : Simple Pulley

✓ connected Particle

✓ $F = ma$

✓ $\boxed{\text{string}}$ → light, → inextensible

✓ Pulley → fixed, → smooth

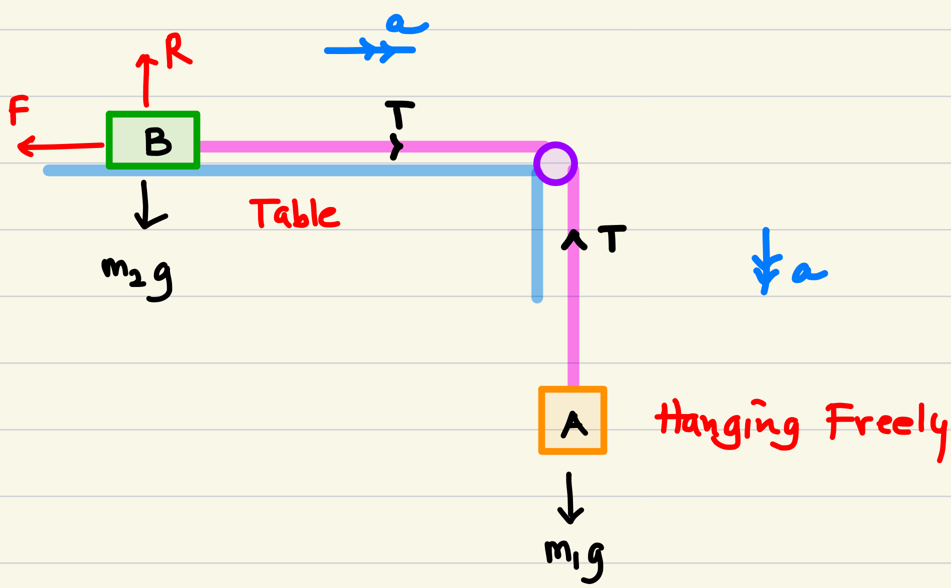
Case II : Two Particles + Table

$m_A \rightarrow m_1$

$m_B \rightarrow m_2$

"Smooth" ✓

"Rough" → $F_{max} = \mu R$
→ Const. Resist. Force ✓



Smooth Surface :

$$m_1 g - T = m_1 a$$

$$T - 0 = m_2 a$$

$$m_1 g = m_1 a + m_2 a$$

$$m_1 g = (m_1 + m_2) a$$

Rough Surface :

$$m_1 g - T = m_1 a$$

$$T - F = m_2 a$$

$$m_1 g - F = m_1 a + m_2 a$$

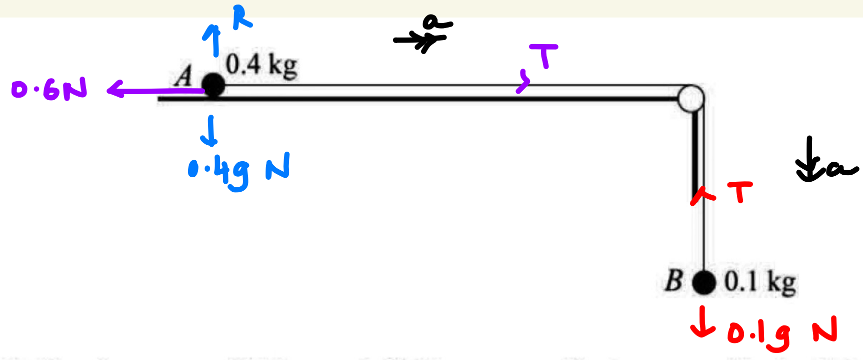
$$= (m_1 + m_2) a$$

Topic 03: Newtons Laws of Motion

(A) Pulleys

CASE II : PP Topical Q's

Question 1:



$R = 0.4g$
$= 4N$

Particles A and B, of masses 0.4 kg and 0.1 kg respectively, are attached to the ends of a light inextensible string. Particle A is held at rest on a horizontal table with the string passing over a smooth pulley at the edge of the table. Particle B hangs vertically below the pulley (see diagram). The system is released from rest. In the subsequent motion a constant frictional force of magnitude 0.6 N acts on A. Find

(i) the tension in the string, [4]

(ii) the speed of B 1.5 s after it starts to move. [3]

Nov 2003 Paper 4 Q5

(i)

$0.1g - T = 0.1a$ — $\boxed{1}$

$T - 0.6 = 0.4a$ — $\boxed{2}$

$0.1(10) - 0.6 = (0.1 + 0.4)a$

$0.4 = 0.5a$

$a = 4/5$ OR $0.8\ m/s^2$

$T - 0.6 = 0.4(0.8)$

$T - 0.6 = 0.32$

$T = 0.92\ N$

(ii)

$u = 0$

$a = 0.8$

$t = 1.5$

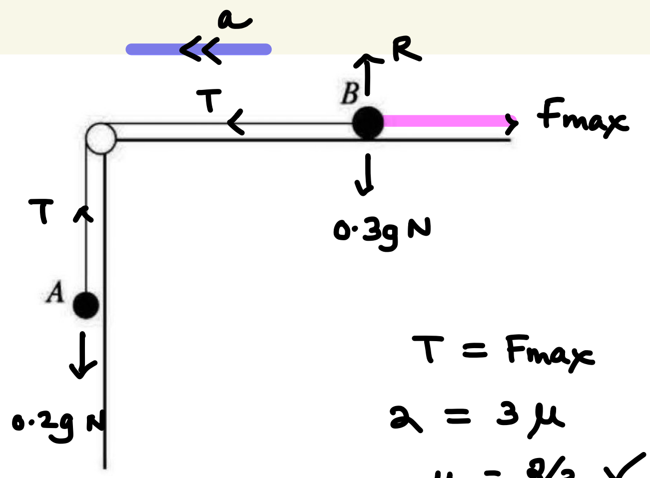
$v = u + at$

$= 0 + (0.8)(1.5)$

$= \frac{4}{5}(1.5)$

$= 1.2\ m/s$

Question 3:



a

$T = 0{\cdot}2(10)$
$= 2N$

$R = 0{\cdot}3(10)$
$R = 3N$

$F_{max} = \mu R$
$= 3\mu$

$T = F_{max}$
$2 = 3\mu$
$\mu = 2/3$ ✓

Particles A and B, of masses 0.2 kg and 0.3 kg respectively, are connected by a light inextensible string. The string passes over a smooth pulley at the edge of a rough horizontal table. Particle A hangs freely and particle B is in contact with the table (see diagram).

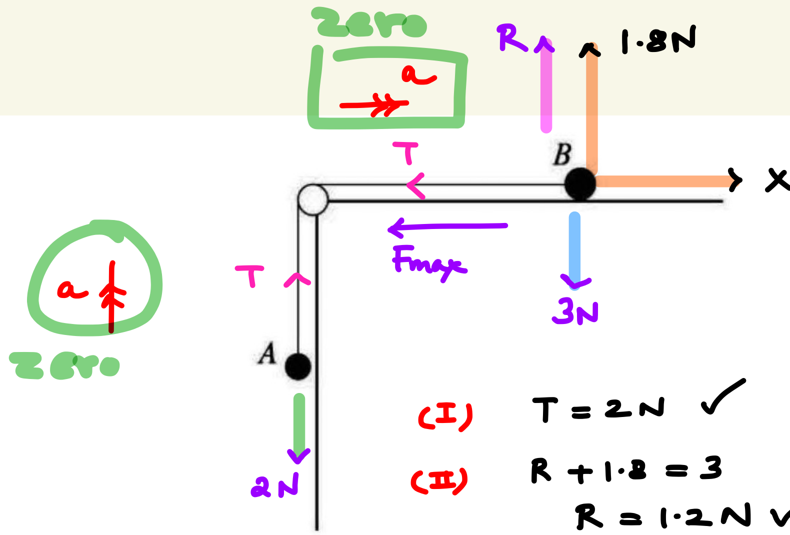
(i) The system is in limiting equilibrium with the string taut and A about to move downwards. Find the coefficient of friction between B and the table. [4]

A force now acts on particle B. This force has a vertical component of 1.8 N upwards and a horizontal component of X N directed away from the pulley.

(ii) The system is now in limiting equilibrium with the string taut and A about to move **upwards**. Find X. [3]

June 2005 Paper 4 Q4

Question 3:



(I) $T = 2N$ ✓

(II) $R + 1.8 = 3$

$R = 1.2N$ ✓

(III) $F_{max} = \mu R$

$= \frac{2}{3}(1.2)$

$= 0.8N$ ✓

(IV) $F_{max} + T = X$

$0.8 + 2 = X$

$\therefore X = 2.8N$

Particles A and B, of masses 0.2 kg and 0.3 kg respectively, are connected by a light inextensible string. The string passes over a smooth pulley at the edge of a rough horizontal table. Particle A hangs freely and particle B is in contact with the table (see diagram).

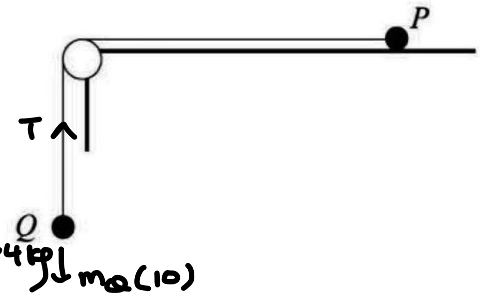
(i) The system is in limiting equilibrium with the string taut and A about to move downwards. Find the coefficient of friction between B and the table. [4]

A force now acts on particle B. This force has a vertical component of 1.8 N upwards and a horizontal component of X N directed away from the pulley.

(ii) The system is now in limiting equilibrium with the string taut and A about to move **upwards**. Find X. [3]

June 2005 Paper 4 Q4

Question 5:



$T = m_Q(10)$

$4 = m_Q(10) \therefore m_Q = 0.4\text{kg}$

Particles P and Q are attached to opposite ends of a light inextensible string. P is at rest on a rough horizontal table. The string passes over a small smooth pulley which is fixed at the edge of the table. Q hangs vertically below the pulley (see diagram). The force exerted on the string by the pulley has magnitude $4\sqrt{2}$ N. The coefficient of friction between P and the table is 0.8.

↳ Force exerted on pulley by the strings

ACTION - REACTION FORCE

(i) Show that the tension in the string is 4 N and state the mass of Q. [2]

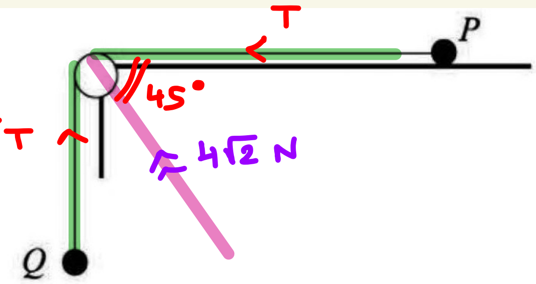
(ii) Given that P is on the point of slipping, find its mass. [2]

A particle of mass 0.1 kg is now attached to Q and the system starts to move.

(iii) Find the tension in the string while the particles are in motion. [4]

June 2006 Paper 4 Q5

Question 5:



$4\sqrt{2}\sin 45° = T$

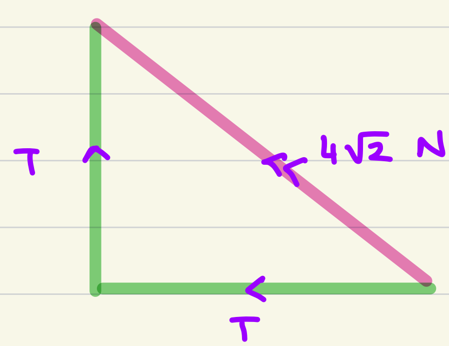
$4\sqrt{2}\left(\frac{1}{\sqrt{2}}\right) = T$

$T = 4$ N

$4\sqrt{2}\cos 45° = T$

$4\sqrt{2}\left(\frac{1}{\sqrt{2}}\right) = T$

$\therefore T = 4$ N.

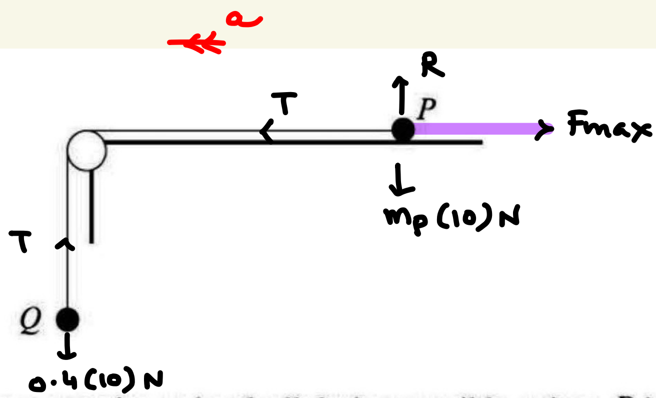


$T^2 + T^2 = (4\sqrt{2})^2$

$2T^2 = 16(2)$

$T^2 = 16 \quad \therefore \underline{T = 4\text{N}}$

Question 5:



i,

$a \downarrow$

$T = 4\text{N}$

$m_Q = 0.4\,\text{kg}$

ii,

$R = m_P(10)$

$F_{max} = \mu R$

$= 0.8(10)(m_P)$

$= 8\,m_P$

$F_{max} = T$

$8\,m_P = 4 \therefore m_P = 0.5\,\text{kg}.$

Particles P and Q are attached to opposite ends of a light inextensible string. P is at rest on a rough horizontal table. The string passes over a small smooth pulley which is fixed at the edge of the table. Q hangs vertically below the pulley (see diagram). The force exerted on the string by the pulley has magnitude $4\sqrt{2}$ N. The coefficient of friction between P and the table is 0.8.

(i) Show that the tension in the string is 4 N and state the mass of Q. [2]

(ii) Given that P is on the point of slipping, find its mass. [2]

A particle of mass 0.1 kg is now attached to Q and the system starts to move. $\Rightarrow$ accelerates

(iii) Find the tension in the string while the particles are in motion. [4]

June 2006 Paper 4 Q5

New mass of Q $\rightarrow$ 0.5 kg

$0.5g - T = 0.5a$

$T - F_{max} = 0.5a$

$0.5g - F_{max} = (0.5 + 0.5)a$

$F_{max} = \mu R$

$= 0.8(0.5)\,10)$

$= 4\,\text{N}$

$0.5(10) - 4 = 1.0\,a$

$a = 1\,\text{m/s}^2$

$T - F_{max} = 0.5a$

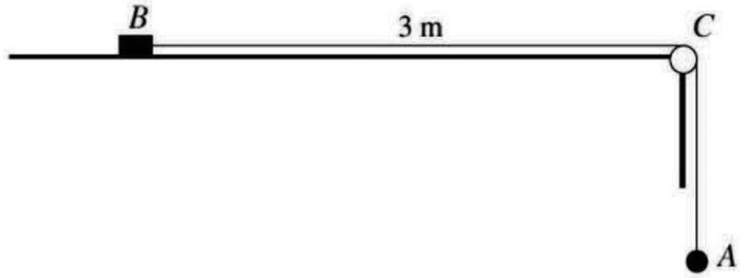
$T - 4 = 0.5(1)$

$T = 4.5\,\text{N}$

Lec # 03

N.L. of motion

Question 8:



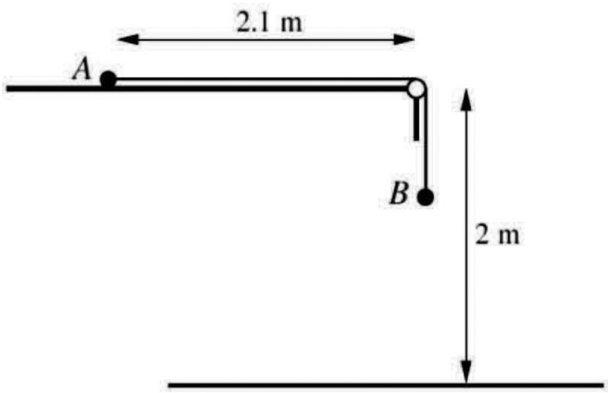
A block B of mass 0.6 kg and a particle A of mass 0.4 kg are attached to opposite ends of a light inextensible string. The block is held at rest on a rough horizontal table, and the coefficient of friction between the block and the table is 0.5. The string passes over a small smooth pulley C at the edge of the table and A hangs in equilibrium vertically below C. The part of the string between B and C is horizontal and the distance BC is 3 m (see diagram). B is released and the system starts to move.

(i) Find the acceleration of B and the tension in the string. [6]

(ii) Find the time taken for B to reach the pulley. [2]

June 2008 Paper 4 Q5

question 13:



Particles A and B, of masses 0.2 kg and 0.45 kg respectively, are connected by a light inextensible string of length 2.8 m. The string passes over a small smooth pulley at the edge of a rough horizontal surface, which is 2 m above the floor. Particle A is held in contact with the surface at a distance of 2.1 m from the pulley and particle B hangs freely (see diagram). The coefficient of friction between A and the surface is 0.3. Particle A is released and the system begins to move.

(i) Find the acceleration of the particles and show that the speed of B immediately before it hits the floor is $3.95\,\mathrm{m\,s^{-1}}$, correct to 3 significant figures. [7]

(ii) Given that B remains on the floor, find the speed with which A reaches the pulley. [4]

June 2010 Paper 41/42 Q6

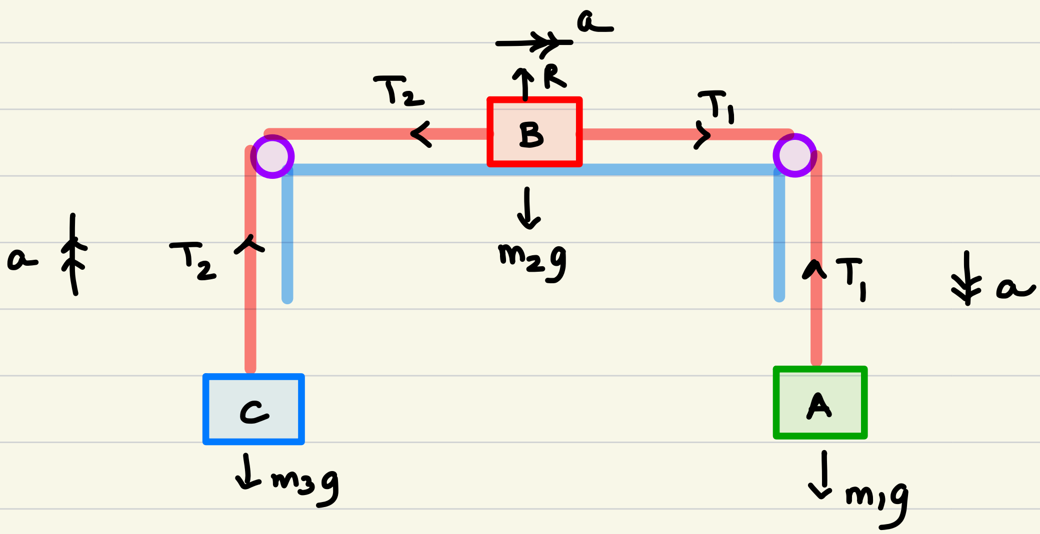
$m_1 > m_3$

Case III : One particle on Table + other Two hanging freely.

$m_A \rightarrow m_1$

$m_B \rightarrow m_2$

$m_C \rightarrow m_3$



SMOOTH TABLE

$$m_1 g - \cancel{T_1} = m_1 a$$

$$\cancel{T_1} - \cancel{T_2} = m_2 a$$

$$\cancel{T_2} - m_3 g = m_3 a$$

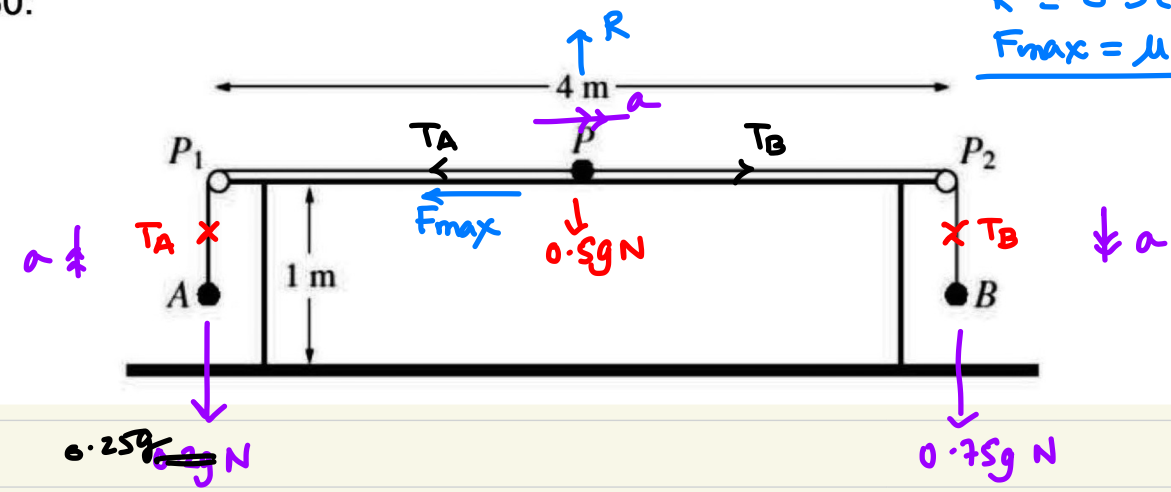
$$m_1 g - m_3 g = (m_1 + m_2 + m_3) a$$

TOPIC 03: NEWTONS LAWS OF MOTION

(A) Pulleys

Question 30:

$R = 0.5(10)$
$F_{max} = \mu R = \cancel{5}\,2\,N$



$$0.75g - T_B = 0.75a$$

$$T_B - T_A - F_{max} = 0.5a$$

$$T_A - 0.25g = 0.25a$$

$$0.75g - \cancel{5}\,2 - 0.25g = 1.5a$$

$$3 = 1.5a$$

$$\therefore a = 2\ m/s^2$$

(i)

$$0.75g - 0.75a = T_B$$
$$7.5 - 0.75a = T_B$$

$$T_A = 0.25a + 0.25g$$
$$= 0.25a + 2.5$$

(iii) $u = 0$

$a = 2$

$s = 0.36$

$$v^2 = 0^2 + 2(2)(0.36)$$

$$v = 1.2\ m/s$$

A light inextensible string of length 5.28 m has particles A and B, of masses 0.25 kg and 0.75 kg respectively, attached to its ends. Another particle P, of mass 0.5 kg, is attached to the mid-point of the string. Two small smooth pulleys P_1 and P_2 are fixed at opposite ends of a rough horizontal table of length 4 m and height 1 m. The string passes over P_1 and P_2 with particle A held at rest vertically below P_1, the string taut and B hanging freely below P_2. Particle P is in contact with the table halfway between P_1 and P_2 (see diagram). The coefficient of friction between P and the table is 0.4. Particle A is released and the system starts to move with constant acceleration of magnitude a m s^{-2}. The tension in the part AP of the string is T_A N and the tension in the part PB of the string is T_B N.

(i) Find T_A and T_B in terms of a. [3]

(ii) Show by considering the motion of P that $a = 2$. [3]

(iii) Find the speed of the particles immediately before B reaches the floor. [2]

* (iv) Find the deceleration of P immediately after B reaches the floor. [2]

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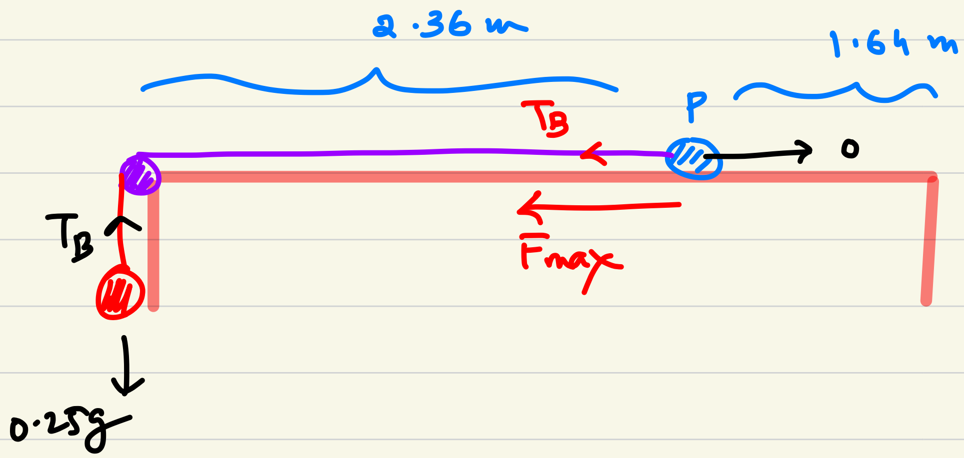


$$0 - T_B - F_{max} = 0.5a$$

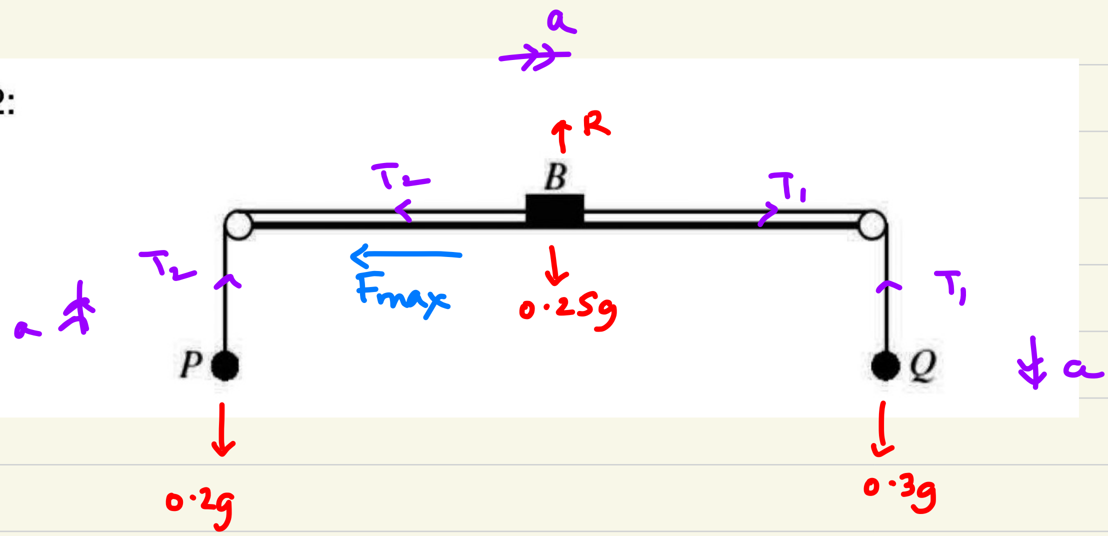
$$T_B - 0.25g = 0.25a$$

$$-2 - (2.5) = 0.75a$$

$$-4.5 = 0.75a$$

$$\underline{\underline{a = -6 \text{ m/s}^2}}$$

Question 32:



$$F_{max} = \mu R$$
$$= \mu(0.25)(10)$$
$$= 2.5\mu$$
$$= \tfrac{5}{2}\mu$$

$$0.3g - T = 0.3(0)$$

$$T_1 - T_2 - F_{max} = 0.25(0)$$

$$T_2 - 0.2g = 0.2(0)$$

$$0.3g - F_{max} - 0.2g = 0$$

$$1 - F_{max} = 0 \qquad \therefore F_{max} = 1$$

$$\frac{5}{2}\mu = 1 \qquad \therefore \mu = 2/5 \checkmark$$

A small block B of mass 0.25 kg is attached to the mid-point of a light inextensible string. Particles P and Q, of masses 0.2 kg and 0.3 kg respectively, are attached to the ends of the string. The string passes over two smooth pulleys fixed at opposite sides of a rough table, with B resting in limiting equilibrium on the table between the pulleys and particles P and Q and block B are in the same vertical plane (see diagram).

(i) Find the coefficient of friction between B and the table. [3]

Q is now removed so that P and B begin to move.

(ii) Find the acceleration of P and the tension in the part PB of the string. [6]

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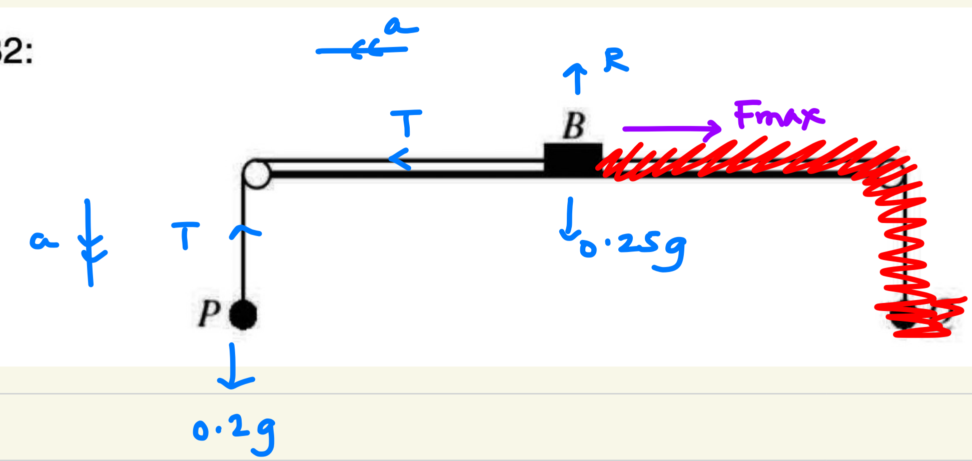
Question 32:



$F_{max} = 1$.

$$0.2g - T = 0.2a$$

Lec 04

$$T - F_{max} = 0.25a$$

$$0.2(10) - (1) = 0.45a$$

$$a = \frac{1}{0.45} = 2.22 \text{ m/s}^2$$

$$T - F_{max} = 0.25a$$

$$T - 1 = 0.25(2.22)$$

$$T =$$

Question 43:

Particles P and Q, of masses 7 kg and 3 kg respectively, are attached to the two ends of a light inextensible string. The string passes over two small smooth pulleys attached to the two ends of a horizontal table. The two particles hang vertically below the two pulleys. The two particles are both initially at rest, 0.5 m below the level of the table, and 0.4 m above the horizontal floor (see diagram).

(i) Find the acceleration of the particles and the speed of P immediately before it reaches the floor. [4]

(ii) Determine whether Q comes to instantaneous rest before it reaches the pulley directly above it. [2]